WORLDMIRROR: AGENTIC REAL-TO-SIM GENERATION OF INTERACTIVE WORLDS FROM IN-THE-WILD ROBOT DEMONSTRATIONS

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ABSTRACT

Existing approaches for robot learning in simulation often require substantial manual effort to construct scenes and specify tasks, limiting training scalability and data diversity. Meanwhile, in-the-wild robot demonstrations naturally provide diverse and realistic observations of real-world interactions. Leveraging these demonstrations for simulation requires both high-fidelity visual reconstruction and physically consistent interactions. General-purpose agents offer flexible adaptation to diverse scenarios and strong reasoning about 3D spatial relationships, while specialized generative models provide high-fidelity reconstruction of scene appearance and geometry. We introduce **WorldMirror**, an agentic real-to-sim framework that constructs interactive digital twins from in-the-wild robot demonstrations. Our central idea is to formulate real-to-sim construction as the iterative refinement of an executable world program. A coding agent decomposes the observed scene, plans reconstruction steps, and orchestrates specialized image and 3D generation tools, refining the program through verification feedback. A camera-pose-anchored scene representation organizes the world into static visual context, support geometry, and editable interactive assets while preserving spatial relationships among the scene, objects, and robot. The agent uses this representation to diagnose and correct reconstruction errors, enabling physically consistent interaction and trajectory replay. Experiments on DROID demonstrate that WorldMirror can reconstruct visually faithful and geometrically consistent interactive worlds while preserving the physical validity of recorded robot trajectories.

1 INTRODUCTION

Scaling generalist robot policies requires diverse training data across a wide range of real-world scenarios (Physical Intelligence, 2025; Black et al., 2024; Wang et al., 2026). Yet collecting real-world demonstrations and repeatedly testing policies across environments remain costly and labor-intensive. Simulation offers a complementary route to scalable data generation and controlled, repeatable evaluation (Li et al., 2024; Jain et al., 2025). Realizing this potential requires interactive environments that reproduce the appearance and physical interactions of real-world deployment settings.

For simulation to serve these purposes, scene geometry, object placement, camera viewpoints, and robot configuration must closely match the real setup. Even small errors in scale or contact geometry can alter task outcomes. Recent real-to-sim systems have advanced scene reconstruction and its use in robot learning and evaluation (Zhao et al., 2025; Ranawaka et al., 2026; Jain et al., 2025; Zhang et al., 2026; Dai et al., 2024; Torne et al., 2024). However, real-to-sim construction involves coupled visual, geometric, and physical constraints that an initial feed-forward reconstruction may not fully satisfy. Meanwhile, general-purpose agents alone may not produce the fine-grained geometry and photorealistic appearance required for faithful scene reconstruction. Different reconstruction errors require different forms of correction. A misplaced object may require layout adjustment, whereas a missing object may require revisiting scene decomposition and asset reconstruction. Unsuitable collision geometry may instead require revising an asset's physical representation. Addressing these errors calls for an adaptive and iterative correction process that selects and revisits reconstruction



Figure 1: WorldMirror enables simulation-ready digital twins with strong realism and interaction fidelity. While GPT-6 Astra benefits from large-scale priors for geometry and layout understanding, WorldMirror achieves better visual realism and physical executability through agentic reconstruction and simulator grounding.

steps based on the current world state and verification feedback, rather than simply following a fixed sequence of tool calls. These challenges motivate *agentic world modeling*, in which a coding agent coordinates specialized reconstruction tools and uses verification feedback to refine an executable world.

We introduce **WorldMirror**, an agentic real-to-sim framework that converts robot videos and their associated trajectories into interactive digital twins. At its core, a coding agent decomposes the reconstruction objective into subtasks and coordinates specialized reconstruction tools to construct an executable world program. Using the current world state and verification feedback, the agent determines which components to revise and which tools to invoke next, updating generated assets and assembly code accordingly. A camera-pose-anchored scene representation organizes the reconstructed world into static visual context, support geometry, and editable interactive assets. It places the scene, objects, and robot in a shared coordinate system, providing a consistent geometric reference for replaying recorded robot trajectories without modification. Table 1 compares WorldMirror with representative real-to-sim and agentic world construction systems.

We evaluate WorldMirror on DROID demonstrations across visual quality, layout fidelity, and motion and interaction fidelity (DROID Team et al., 2024). The results show competitive visual reconstruction quality and improved motion and interaction fidelity during replay compared with the evaluated baselines.

Our contributions are:

- We propose **WorldMirror**, an agentic real-to-sim framework that constructs interactive digital twins through adaptive tool orchestration and verification-guided refinement of executable world programs.

- We introduce a camera-pose-anchored scene representation that organizes reconstructed worlds into static visual context, support geometry, and editable interactive assets, enabling spatially consistent replay of recorded robot trajectories.

Table 1: Capability comparison of representative real-to-sim and agentic world construction systems, covering scene reconstruction, robot-world alignment, and agentic construction.

Capability	RoLA	RoboSnap	SimFoundry	SAGE	Astra	Agentic Real2Sim	WorldMirror
Scene Reconstruction							
Scene Appearance	✓	✓	✓	✓	✗	✗	✓
Interactive Assets	✓	✓	✓	✓	✓	✓	✓
Robot–World Alignment							
Camera Calibration	✗	✓	✓	✗	✓	✓	✓
Demonstration Replay	✗	✓	✗	✗	✓	✓	✓
Agentic Construction							
Tool Orchestration	✗	✗	✗	✓	✓	✓	✓
Agent-based Verification	✗	✗	✗	✓	✓	✓	✓
Feedback-guided Refinement	✗	✓	✓	✓	✓	✓	✓

- We evaluate WorldMirror on DROID across visual quality, layout fidelity, and motion and interaction fidelity, demonstrating competitive reconstruction quality and improved replay fidelity on the evaluated demonstrations.

2 Related Work

Real-to-sim world construction. Constructing interactive digital twins requires recovering both the appearance of a real scene and the physical structure needed to support robot interaction. Existing systems approach this problem through different reconstruction strategies: RialTo builds scanned digital twins for policy improvement (Torne et al., 2024), while RoLA constructs interactive environments from individual images (Zhao et al., 2025). PolaRiS further develops scalable real-to-sim reconstruction for generalist robot policy evaluation, using neural reconstruction to turn short video scans of real-world scenes into interactive simulation environments (Jain et al., 2025). Several methods further integrate reconstruction with explicit alignment and physical refinement. Other approaches explicitly incorporate geometric alignment and physical refinement. SimFoundry combines video-based reconstruction, pose alignment, and physics annotation, and generates digital cousins for robot evaluation and training (Ranawaka et al., 2026). RoboSnap separates interactive geometry from Gaussian visual context and refines scene configurations through geometric and physics optimization (Zhang et al., 2026), while TwinAligner jointly addresses visual alignment and rigid-body system identification (Fan et al., 2025). These approaches establish visual reconstruction, physical geometry, and robot-world alignment as important components of interactive real-to-sim systems. Building on these advances, WorldMirror organizes reconstruction and alignment within an executable world program, where specialized agents coordinate component-level updates guided by verification feedback.

Agentic 3D Modeling. Recent advances in multimodal agents have enabled programmatic construction and manipulation of 3D content. GPT-6 Astra (OpenAI, 2026) demonstrates the use of general-purpose agents for constructing complex 3D worlds, while Articraft (Zhou et al., 2026) and Procedura (Lin et al., 2026) explore agent-driven asset generation and procedural modeling.

Agentic approaches have also incorporated specialized generation tools and verification mechanisms for scene construction. SAGE (Xia et al., 2026) coordinates scene and object generators with critics to iteratively construct simulation-ready environments. LiteReality (**?**) reconstructs RGB-D scans into graphics-ready scenes with structured layouts, retrieved object assets, realistic materials, and basic physical properties. These works illustrate different approaches to combining structured scene representations, specialized reconstruction tools, and agentic generation for 3D world construction.

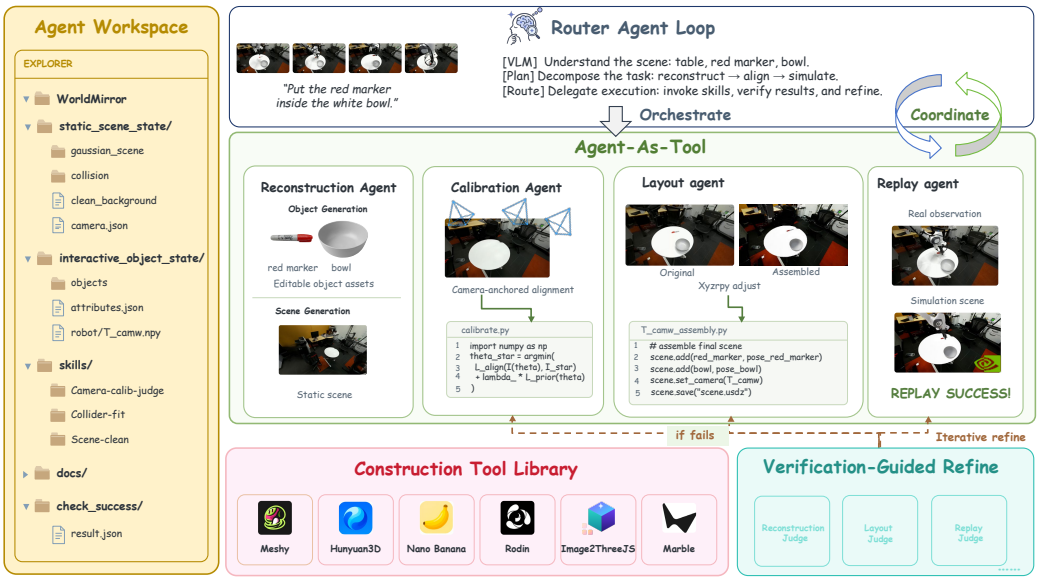


Figure 2: **WorldMirror framework.** A coding agent maintains an editable world, invokes scene and object generation tools, and aligns camera, assets, and robot through assembly code. Rendered observations and execution feedback guide targeted revisions. The resulting world supports trajectory replay, policy evaluation, and simulation-data generation.

More closely related to our setting, Agentic Real2Sim (**?**) combines agentic reconstruction, scene calibration, and simulator-guided refinement to reconstruct recorded robot interactions. WorldMirror similarly addresses agentic reconstruction of interactive environments, with an emphasis on hierarchical construction over a shared, editable world state. Its router coordinates specialized agents across subtasks, while local verification and refinement support iterative updates to reconstructed components. The resulting world is assembled under a camera-anchored coordinate system to support replay of recorded robot trajectories.

Simulation for Robot Learning Simulation environments provide an important foundation for scalable robot learning. RLBench offers a diverse suite of language-conditioned manipulation tasks with reusable simulation environments and demonstrations (James et al., 2020), while robosuite provides a modular framework for constructing and evaluating robot manipulation tasks (Zhu et al., 2020). ManiSkill further scales simulation-based learning with diverse manipulation tasks, object assets, and large-scale demonstrations (Gu et al., 2023). SIMPLER demonstrates the use of simulation as a scalable and reproducible complement to real-world evaluation, providing simulated environments for common real robot setups (Li et al., 2024). MimicGen adapts a small set of human demonstrations to diverse simulated scenes and object configurations (Mandlekar et al., 2023), while RoboTwin provides large-scale simulation environments and task variations for robot manipulation learning (Chen et al., 2025). These approaches demonstrate the potential of simulation for scalable robot learning by providing increasingly diverse simulation environments and large-scale demonstrations. WorldMirror complements these efforts by combining agentic real-to-sim reconstruction with in-the-wild robot demonstrations to construct executable interactive worlds.

3 Method

3.1 Problem Formulation

We study the problem of constructing simulation-ready digital twins from in-the-wild robot manipulation demonstrations. Given an RGB video sequence $\mathcal{V}$ and the corresponding robot trajectory $\mathcal{T}$, WorldMirror constructs a digital twin:

$$\hat{\mathcal{S}} = \mathcal{A}(\mathcal{V}, \mathcal{T}), \tag{1}$$

where $\mathcal{A}$ denotes the agentic reconstruction process. The resulting digital twin consists of a reconstructed environment, editable interactive objects, a calibrated camera configuration, and the robot setup required for trajectory replay.

A simulation-ready digital twin should satisfy three requirements. First, it should preserve the visual appearance and spatial organization of the observed environment. Second, task-relevant objects should be represented as editable simulation entities with appropriate geometry, physical properties, and interaction semantics. Third, the scene, objects, and robot should share a consistent coordinate system that supports replay of the recorded trajectory without modification.

WorldMirror constructs and refines the digital twin through an executable world program that assembles the reconstructed components. Throughout construction, the agent maintains an explicit world state containing reconstructed entities, spatial relationships, and intermediate verification results. This state guides subsequent construction operations and revisions to generated assets and assembly code.

3.2 Structured World State Memory

WorldMirror maintains a structured world state that represents the evolving reconstruction and serves as shared memory across specialized agents. The state explicitly organizes reconstructed entities, their spatial relationships, and simulation-relevant attributes, providing persistent context for subsequent construction and verification. As construction progresses, intermediate reconstruction results and verification feedback are incorporated into the shared state, enabling continuous updates and feedback-driven refinement.

Formally, we represent the world state at construction step t as

$$\mathcal{M}_t = (\mathcal{S}_t, \mathcal{O}_t, \mathcal{R}_t), \tag{2}$$

where $\mathcal{S}_t$ denotes the static scene state, $\mathcal{O}_t$ denotes the interactive object state, and $\mathcal{R}_t$ captures spatial relationships among reconstructed entities. The updated state provides a shared context for subsequent agent decisions and skill execution.

Static scene state. The static scene state maintains the persistent environmental context of the target task, including fixed structures such as tables, walls, and other background elements. We represent scene appearance using a Gaussian representation and model physical support through corresponding collision geometry. Together, these components provide the visual context and geometric support required for interactive object placement and physical simulation.

Interactive object state. The interactive object state maintains task-relevant entities involved in robot manipulation. Each object is represented by its geometry, pose, physical properties, and task-relevant interaction attributes, enabling its instantiation as an editable simulation entity. These attributes can be selectively updated during construction and refinement to address discrepancies identified through verification.

3.3 Agentic World Construction

Router-based Orchestration. WorldMirror employs a router agent to coordinate the construction of an executable world program. The router reasons over the robot demonstration, current world state, and verification feedback to decompose the reconstruction objective into subtasks and select appropriate specialized agents. As construction progresses, it determines which components to revisit and which operations to perform, adapting the execution sequence to intermediate reconstruction results.

Agent-as-Tool Delegation. WorldMirror adopts an agent-as-tool paradigm, where specialized agents are exposed as callable capabilities to the router agent. Unlike individual construction skills that execute specific operations, specialized agents handle complex subtasks requiring multi-step reasoning and coordination of multiple skills. Operating on the shared world state, these agents provide dedicated capabilities for camera calibration, spatial layout reasoning, asset reconstruction, and trajectory replay. The router coordinates subtasks, while specialized agents manage skill execution within their respective domains.

Construction Skill Library. WorldMirror maintains a library of reusable construction skills that encapsulate specialized perception, generation, and simulation operations. These skills are invoked by specialized agents to execute construction and refinement subtasks, including:

- **Camera calibration.** This skill estimates camera parameters to align the reconstructed world with real observations. We initialize camera correspondence through DINO-based image matching (Oquab et al., 2023) and refine the estimation using structural edge alignment (Von Gioi et al., 2012).
- **Spatial layout.** This skill estimates and refines the placement of reconstructed entities within the scene. It adjusts object poses to align generated assets with real observations while maintaining spatial consistency among the static scene, interactive objects, and robot.
- **Asset construction.** This skill generates simulation-ready visual and physical assets from the reconstructed world state. For static environments, it leverages generative scene reconstruction to recover visual context and supporting geometry, while task-relevant objects are reconstructed as editable assets using appropriate generation methods, such as Meshy for rigid objects and Articraft (Zhou et al., 2026) for articulated objects.

3.4 Verification-driven Refinement

Iterative Verification. Throughout world construction, WorldMirror iteratively verifies the reconstructed world against the original demonstration. Specialized judging agents assess reconstruction, spatial layout, and trajectory replay using both semantic reasoning and task-specific verification skills. These agents combine quantitative measurements with contextual judgments to identify discrepancies that cannot be fully captured by predefined metrics alone. The resulting verification feedback is incorporated into the shared world state to guide subsequent refinement.

Closed-loop Refinement. Verification feedback is incorporated into the shared world state and provided to the corresponding specialized agents for targeted refinement. Within their respective subtasks, these agents interpret the feedback and invoke appropriate construction skills to revise affected components, including assets, spatial configurations, and assembly code. When discrepancies involve multiple subtasks, specialized agents communicate with the router to coordinate further refinement. The updated world is subsequently re-evaluated, enabling iterative refinement at both the subtask and world levels.

3.5 Camera Pose-Anchored Coordinate System

After camera calibration, WorldMirror establishes a camera-anchored coordinate system that serves as the geometric reference for the reconstructed digital twin. Unlike independently reconstructed components that may suffer from inconsistent poses, all scene elements, interactive objects, and robot embodiments are expressed within the same viewer-world frame, preserving the spatial correspondence observed in the original demonstration.

Let

$$T_{\mathrm{camw}} = \begin{bmatrix} R & t \\ \mathbf{0}^\top & 1 \end{bmatrix} \in SE(3)$$

denote the transformation from the camera frame to the unified world frame. All reconstructed components are transformed into this reference frame through rigid transformations:

$$X_w = T_{\mathrm{camw}} X_c,$$

where X_c and X_w denote points represented in the camera and viewer-world coordinate systems, respectively.

This camera-anchored formulation enables consistent assembly of the reconstructed environment, editable object assets, and robot trajectory. Specifically, generated assets are placed according to their observed spatial correspondence, while the recorded robot trajectory is expressed in the shared coordinate frame without motion replanning or trajectory retargeting. The resulting digital twin preserves visual alignment and geometric consistency, providing the foundation for physically consistent interaction replay in simulation.

Table 2: Visual quality comparison on 10 DROID scenes. Higher is better for ↑ metrics, while lower is better for ↓ metrics.

Type	Method	PSNR ↑	SSIM ↑	LPIPS ↓	DISTS ↓	Aesthetic ↑
Open-source	RoLA (Zhao et al., 2025)	13.53	**0.587**	0.506	0.206	0.480
	RoboSnap (Zhang et al., 2026)	10.30	0.483	0.641	0.209	0.532
Proprietary	GPT-6 Astra	11.90	0.515	0.606	0.299	0.472
Ours	**WorldMirror**	**13.57**	0.570	**0.427**	**0.168**	**0.570**

4 EXPERIMENTS

We evaluate WorldMirror on in-the-wild robot manipulation demonstrations from DROID (DROID Team et al., 2024). Given recorded RGB observations and robot trajectories, WorldMirror automatically constructs simulation-ready digital twins through its agentic construction framework. We evaluate the resulting digital twins in terms of visual fidelity, spatial consistency, and interaction consistency against the original demonstrations.

4.1 EXPERIMENTAL SETUP

Comparisons. Our reconstruction baselines are RoLA (Zhao et al., 2025), RoboSnap (Zhang et al., 2026), SimFoundry (Ranawaka et al., 2026), and GPT-6 Astra (OpenAI, 2026). Each table lists the methods evaluated for the corresponding metrics.

Metrics. We evaluate digital twin fidelity at three levels: visual fidelity, spatial fidelity, and replay fidelity. Visual fidelity is measured using PSNR, SSIM, LPIPS, DISTS, and aesthetic scores; spatial fidelity is evaluated using mask IoU, position error, and relative vector and distance errors; and replay fidelity is measured using object-motion errors, contact F1, and success rate.

4.2 MIRRORING REAL-WORLD APPEARANCE

We compare first-frame simulation renders with real exterior-camera images on ten DROID scenes. Table 2 reports quantitative results, and Figure 3 provides qualitative comparisons of reconstructed views. We further visualize the wrist-view reconstruction timeline in Figure 4, showing the correspondence between the observed robot demonstration and the reconstructed simulation sequence. WorldMirror achieves the lowest LPIPS and DISTS, together with competitive PSNR and aesthetic scores among evaluated methods. These results support improved visual correspondence on the evaluated views, particularly under perceptual metrics.

4.3 PRESERVING REAL-WORLD SPATIAL STRUCTURE

To assess whether reconstructed scenes preserve real-world spatial structure, we evaluate object placement and inter-object relationships on a shared set of 17 task-relevant objects and eight object pairs from DROID. We report mask IoU, position error, and relative vector and distance errors, with the 3D metrics computed from visible-surface centroids. As shown in Table 3, WorldMirror outperforms the evaluated open-source baselines across all four metrics. Compared with GPT-6 Astra, however, it has lower mask IoU and higher position and relative vector errors. WorldMirror achieves the lowest relative distance error, indicating better preservation of pairwise distances on the evaluated scenes, but not a consistent advantage across spatial metrics.

4.4 PRESERVING REAL-WORLD MOTION AND INTERACTION

To evaluate whether reconstructed digital twins preserve the physical conditions required for real-world manipulation, we replay the original DROID robot trajectories in the reconstructed environments using Isaac Sim without any trajectory modification or additional motion retargeting. Unlike approaches that require re-planning or inverse kinematics based adaptation to transfer demonstra-

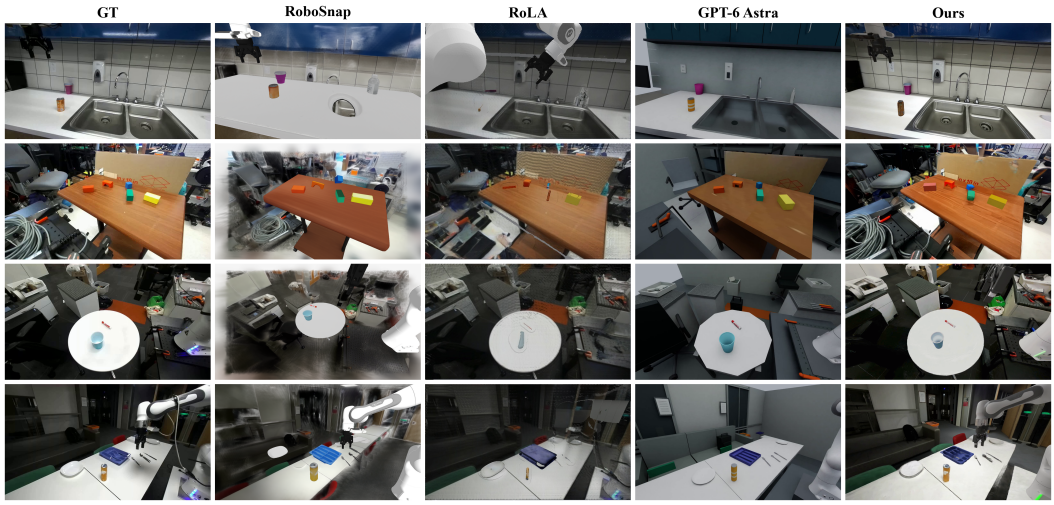


Figure 3: Qualitative RGB comparison with existing baselines on DROID scenes. WorldMirror produces more faithful visual reconstructions with improved appearance consistency and scene details compared with open-source and proprietary baselines.

Table 3: Object layout fidelity comparison. Higher is better for ↑ metrics, while lower is better for ↓ metrics.

Category	Method	IoU ↑	Pos. Err. ↓	Rel. Vec. ↓	Rel. Dist. ↓
Open-source	RoLA (Zhao et al., 2025)	0.232	30.65	0.814	0.542
	RoboSnap (Zhang et al., 2026)	0.313	27.79	0.644	0.583
	SimFoundry (Ranawaka et al., 2026)	0.169	19.54	0.627	0.507
Proprietary	GPT-6 Astra (OpenAI, 2026)	**0.598**	**8.72**	**0.552**	0.331
Ours	**WorldMirror**	0.469	9.66	0.557	**0.219**

tions into simulation, WorldMirror directly executes the recorded robot trajectories in the reconstructed coordinate system, providing a stricter evaluation of geometric and physical consistency.

We assess motion consistency using displacement, final-position, direction, and DTW errors, and evaluate interaction fidelity using contact F1 and replay success rate.

As shown in Table 4, WorldMirror achieves the lowest errors across all four motion metrics and higher contact F1 than GPT-6 Astra. It achieves a replay success rate of 100% on the evaluated demonstrations, compared with 40% for GPT-6 Astra. These results demonstrate that the reconstructed digital twins preserve the spatial and physical consistency required for direct trajectory replay and robot-environment interaction.

4.5 Ablation

5 Conclusion

We introduced **WorldMirror**, an agentic real-to-sim framework that constructs simulation-ready digital twins from in-the-wild robot demonstrations. Unlike conventional reconstruction pipelines that rely on fixed sequences of specialized modules, WorldMirror formulates digital twin construction as an adaptive process in which a router agent coordinates specialized agents and skills, maintains a structured world state, and refines the reconstruction through verification-driven feedback.

Experiments on DROID demonstrate that WorldMirror preserves visual appearance, spatial structure, and motion-interaction consistency of real-world demonstrations. By maintaining a camera-pose anchored coordinate system, WorldMirror enables direct trajectory replay in reconstructed en-



Figure 4: Real-to-sim wrist-view comparison over time. The reconstructed digital twin closely matches the real robot observations throughout the manipulation trajectory, demonstrating consistent visual alignment during replay.

Table 4: Motion replay and physical interaction evaluation of reconstructed digital twins on DROID demonstrations. Motion metrics measure trajectory consistency, while interaction metrics evaluate contact fidelity and replay success in Isaac Sim. F1 denotes contact F1 score, and SR denotes replay success rate.

	Motion Replay				Physical Interaction	
Method	Disp. (cm)↓	Final (cm)↓	Dir. (°)↓	DTW (cm)↓	F1↑	SR↑
RoLA (Zhao et al., 2025)	24.70	40.80	53.00	10.30	–	–
GPT-6 Astra (OpenAI, 2026)	11.17	21.98	81.30	10.82	0.653	40%
WorldMirror	**1.79**	**5.83**	**10.90**	**1.64**	**0.873**	**100%**

vironments without trajectory modification, providing simulation-ready digital twins for robot interaction evaluation.

We hope WorldMirror serves as a step toward scalable robot learning and evaluation by reducing the manual effort required to transform diverse real-world experiences into reliable simulation environments.

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6 APPENDIX